Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

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***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a low-dimensional latent space under distribution shift, alongside an Operator-Entropy variant (OpEntropy JEPA) combining a linear Koopman transition operator, spectral entropy and shifted logarithmic Bregman barrier regularizers, coordinate saliency gating, and a multiscale dyadic frequency front-end. Across 33 continuous telemetry streams from seven physical datasets under chronological held-out Streaming Peaks-Over-Threshold calibration, our primary finding is relative false-alarm suppression rather than universal detection superiority or exact attainment of the 0.1% nominal design risk. Latent predictive scoring reduces held-out false-positive rates three- to four-fold under retrospective buffered attribution (7.4% to 10.1% versus 30.0% to 30.7% for TimesNet and TranAD) and eight- to twenty-fold under strictly causal streaming (1.1% to 2.8% versus 22.2% to 24.3%), while point F1 and precision-recall ranking remain comparable to reconstructive baselines. Controlled ablations isolate two distinct mechanisms: LayerNorm bottlenecks compress residual kurtosis, whereas learned latent projections halve calibration-to-holdout distribution drift relative to normalized observation forecasters. On a fixed Koopman backbone, equivalence testing confirms that spectral entropy and barrier terms preserve full operator rank without altering point F1 within a 0.01 to 0.03 margin, while the OpEntropy transition architecture shortens detection delay relative to unconstrained TS-JEPA. Static thresholding still overshoots its 0.1% design target by 15- to 64-fold out of sample, driven primarily by inter-block distribution shift and serial dependence.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments such as NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3]. Operational monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous anomaly scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains ambient sensor jitter and channel-scale disparities, observation-space residuals exhibit heavy-tailed excess kurtosis ($\gamma_2 > 8$) and strong serial clustering. When paired with automated extreme-value thresholding such as Streaming Peaks-Over-Threshold (SPOT) [7] at nominal design risk $q = 10^{-3}$ (0.1%), these heavy-tailed residuals suffer severe calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: To suppress false-alarm flooding without synthetic anomaly injection [8], we study how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a low-dimensional latent space [9] under distribution shift, and formulate an *Operator-Entropy JEPA* (OpEntropy JEPA) transition architecture regularized by Koopman spectral entropy and a shifted Bregman energy barrier. Rather than claiming that physics-inspired regularizers drive anomaly detection accuracy, we evaluate them strictly as algebraic conditioning priors on abstract latent coordinates. Our evaluation separates three operational questions: (i) relative false-alarm suppression under automated calibration versus exact attainment of the nominal 0.1% risk target, (ii) discriminative detection across operating points and fault classes, and (iii) structural score-emission latency versus empirical fault-detection delay.

A. Contributions

This study makes four focused contributions:

1) **Primary Mechanism Isolation (Compression vs. Drift):** Across 12 controlled backbone arms on all 19 non-GHL streams (Section VI-E, Fig. 4(a), Table I(e)), adding a D=32 LayerNorm bottleneck to an observation forecaster compresses calibration kurtosis (γ_2 : $6.10 \rightarrow 0.91$) and halves causal false-positive rates ($24.7\% \rightarrow 11.7\%$), while

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learned latent projection in TS-JEPA and OpEntropy JEPA halves calibration-to-holdout Kolmogorov–Smirnov drift (D_{KS} : $0.376 \rightarrow 0.209$–0.214) and lowers causal false alarms to 0.9%–2.8%.

2) **Relative False-Alarm Suppression vs. Detection Limits:** Across 33 streams from seven datasets (Table II), latent scoring reduces relative held-out false-alarm rates three- to four-fold under buffered attribution (7.4%–10.1% vs. 30.0%–30.7%) and $8\times$–$20\times$ under strictly causal streaming (1.1%–2.8% vs. 22.2%–24.3%). Separating deployable calibration quantiles (q^*) from post-hoc matched realized test-FPR curves (f^*, Fig. 4(b)) shows higher low-FPR causal event recall (0.395–0.431 vs. 0.217 at $f^*{=}0.01$), whereas Point-F1, PR-AUC, and buffered recall match reconstructors and quiet flatlines remain weakly detected (0.02–0.07 recall).
3) **Operator-Entropy Conditioning and Equivalence Bounds:** Proposition 1 establishes non-triviality conditions for the Koopman transition operator under finite-ϵ and singular energy barriers. While the OpEntropy transition architecture cuts buffered detection delay from 110.3 to 79.5 steps ($127.0 \rightarrow 58.2$ causal), fixed-backbone Two One-Sided Tests ($\lambda_{\text{ent}}{=}0$ control, Table I(a)) confirm Point-F1 equivalence within $\Delta = \pm 0.03$ and down to ± 0.010, isolating the spectral regularizer as a conditioning constraint rather than a detection driver.
4) **Operational Calibration Gap and Refitted Permutation Audit:** Static SPOT misses its nominal 0.1% target out of sample (1.51%–1.95% stream-macro causal and 5.54%–6.38% buffered FPR across 33 streams for JEPAs, or 15–$20\times$ and 55–$64\times$ overshoot vs. 142–$222\times$ for reconstructors, Table IV). A three-stage refitted permutation decomposition (Table IV(c)) estimates that inter-block shift accounts for 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors) and serial dependence for 29.1%–35.3%.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly operate in raw observation space by training a network to forecast future sensor readings or reconstruct input windows, flagging timesteps where the point-level observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive LSTMs with non-parametric dynamic thresholding [3] model sequential dependencies directly on raw spacecraft telemetry channels. More recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [4] contrasts prior Gaussian kernel associations against series-level self-attention to amplify association discrepancies; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods so 2D convolutional kernels can capture intra- and inter-period variations.

Despite strong ranking performance on clean benchmarks, observation-space reconstructors share a structural vulnerability under automated thresholding on noisy multi-sensor arrays. Because point-level losses penalize reconstruction error across every channel equally, the network is forced to model high-frequency stochastic noise, quantization jitter, and background environmental fluctuations. In high-dimensional streams ($K \geq 25$), aggregating raw reconstruction errors across channels accumulates heavy-tailed residual spikes, referred to statistically as *leptokurtic* tails (excess kurtosis $\gamma_2 > 0$), which destabilize extreme-value threshold fitting and inflate false alarms whenever baseline operating conditions shift [10].

Contrastive and Joint-Embedding Predictive Architectures: To avoid reconstructing raw sensor noise, self-supervised representation methods map windows into a compact latent embedding space. Contrastive detectors construct positive and negative pairs: TS2Vec [11] maximizes contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [8] contrasts full window embeddings against context-only embeddings using injected synthetic anomalies; and DCdetector [12] employs dual patch-wise and in-patch attention branches without a reconstruction loss. However, contrastive classifiers depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology, yielding lower calibrated point detection on diverse continuous telemetry (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent predictive scoring across 11 datasets, Table II(d)).

Joint-Embedding Predictive Architectures (JEPAs) [9], [13], [14] eliminate both raw observation decoding and synthetic negative pairs. Rather than reconstructing inputs, a JEPA encodes a historical context window into a latent vector z_{ctx} and predicts the target representation z_{tgt} of a future window produced by a separate target encoder. Because trivial constant embeddings would zero out latent prediction error, non-contrastive architectures prevent representation collapse either through architectural asymmetry (updating the target encoder via an exponential moving average of online weights [15]) or through explicit variance-covariance regularization such as VICReg [16], Barlow Twins [17], and batch whitening [18]. While prior time-series representation studies evaluate embeddings via post-hoc threshold sweeps, how latent predictive residuals behave under strictly unsupervised extreme-value threshold calibration remains unexplored.

Koopman Operator Theory and Physics-Inspired Priors: Multi-sensor telemetry reflects the underlying trajectories of physical dynamical systems [19]. Neural ODEs [20] parameterize continuous velocity fields in observation space, whereas Physics-Informed Neural Networks (PINNs) [21] and Hamiltonian or Lagrangian networks [22], [23] enforce partial differential conservation laws directly on physical state coordinates. However, rigid differential constraints require known governing equations and suffer from stiff competing gradient pathologies on uncalibrated sensor arrays [24].

Koopman operator theory [25], [26] provides a softer dynamical prior: nonlinear dynamics can be lifted into an observable embedding space where time evolution is governed by a linear operator. Data-driven Koopman networks [27]–[29] learn neural embeddings jointly with a finite-dimensional transition matrix $\mathbf{K}$. Here we regularize the singular value spectrum of $\mathbf{K}$ via Von Neumann spectral entropy and a

smooth Bregman energy barrier, maintaining a strict distinction between *physics-inspired* algebraic priors (which shape abstract learned latent coordinates) and *physics-informed* conservation laws, and testing whether operator conditioning yields downstream detection gains or statistical equivalence relative to an unconstrained predictor.

Extreme-Value Threshold Calibration and Evaluation Rigor: Unsupervised streaming deployment requires converting continuous scores into binary alarms using nominal calibration data alone. Streaming Peaks-Over-Threshold (SPOT) [7] fits a Generalized Pareto Distribution (GPD) to score exceedances above a high calibration quantile u to extrapolate a threshold at design risk q. However, classical extreme-value theory assumes independent and identically distributed (i.i.d.) samples. Continuous telemetry violates this assumption through short-range serial correlation, which causes threshold exceedances to occur in multi-step bursts quantified by the Ferro–Segers extremal index [30], and through non-stationary regime drift between calibration and test blocks. Concurrently, recent benchmark audits [10], [31], [32] have shown that "point-adjusted" F1 (which credits an entire anomaly segment if a single timestep triggers an alarm) severely inflates scores; we therefore evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [33], [34], TimeEval [10], and mTSBench [35]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP and MSL entities, with two short-nominal MSL streams reserved; SWAN-SF solar magnetograms [36]); *wearable kinematics* (six Daphnet FoG streams [33]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so $S{=}64$ is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [37], six CICIDS edge network-flow streams, CalIt2 people-flow [34], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [10] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample threshold tuning, each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Stream inclusion in the primary multi-seed protocol required: (i) $N_{\text{cal}} \geq 500$ calibration windows after splitting, and (ii) tractable stride-1 multi-seed runtime. Exactly 33 streams across seven datasets satisfy both criteria (6 Daphnet, $K{=}9$; 14 GHL, $K{=}16$; 4 MSL, $K{=}55$; 6 SMAP, $K{=}25$; Genesis, $K{=}18$; Metro, $K{=}5$; SWAN, $K{=}38$). The remaining 12 streams (CalIt2, $K{=}2$; two Room-Occupancy, $K{=}5$; two short-nominal MSL; GECCO, $K{=}9$; six 72-channel CICIDS flows) appear in operational incident case studies and a secondary 11-dataset, 130-stream single-seed baseline cross-check (Table II(d)); the dyadic front-end ablation (Table I(b)) includes SMD `machine-1-2` ($K{=}38$).

Two external-validity caveats bound interpretation: (i) SMAP and MSL contain documented mislabeled segments and point anomalies [32], so per-dataset scores represent upper bounds on operational detectability; and (ii) because streams within SMAP, MSL, Daphnet, and GHL share hardware and environmental characteristics, streams within a dataset are not independent domains ($N{=}7$ dataset clusters). We therefore report unweighted Dataset-Macro means, between-stream and within-stream seed SDs, leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table II(c)), restricting conclusions strictly to the evaluated entities and sampling regimes.

IV. Methodology: Operator-Entropy Joint-Embedding Predictive Architecture

A. Problem Formulation and Latent Space Prediction

Consider a continuous multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T\times K}$ recording K sensor channels over T timesteps. As illustrated in Fig. 2, a sliding window of total length $W = C + S = 320$ steps advances across the stream at stride 1 and partitions the local series at split index t into two contiguous blocks:

1) **Historical context window ($\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C : t, :] \in \mathbb{R}^{C\times K}$):** The preceding $C = 256$ observations leading up to index t, which establish the current nominal operating regime, phase, and baseline amplitude of the system.
2) **Suspect target horizon ($\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S\times K}$):** The subsequent $S = 64$ observations spanning from t to the online decision timestamp $\tau = t + S - 1$, which are inspected to determine whether the system has departed from its expected trajectory.

Reconstructive detectors (such as TimesNet, TranAD, and Anomaly Transformer) train an autoencoder to reproduce the raw waveform by minimizing pointwise observation error $\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$. Because raw telemetry contains high-frequency sensor noise across all K channels, observation-space residuals exhibit heavy, outlier-prone tails that trigger frequent false alarms.

We instead adopt a Joint-Embedding Predictive Architecture (JEPA) [9] that shifts prediction from raw observation space into a compact 32-dimensional latent space ($D = 32$, Fig. 2, bottom). An online context encoder E_θ compresses $\mathbf{x}_{\text{ctx}}$ into a context vector $z_{\text{ctx}} \in \mathbb{R}^D$, while a separate target encoder E_ϕ compresses $\mathbf{x}_{\text{tgt}}$ into a target vector $z_{\text{tgt}} \in \mathbb{R}^D$. Rather than training E_ϕ by direct gradient descent (which would allow both encoders to map all inputs to a constant vector), the target weights ϕ are updated as an exponential moving average

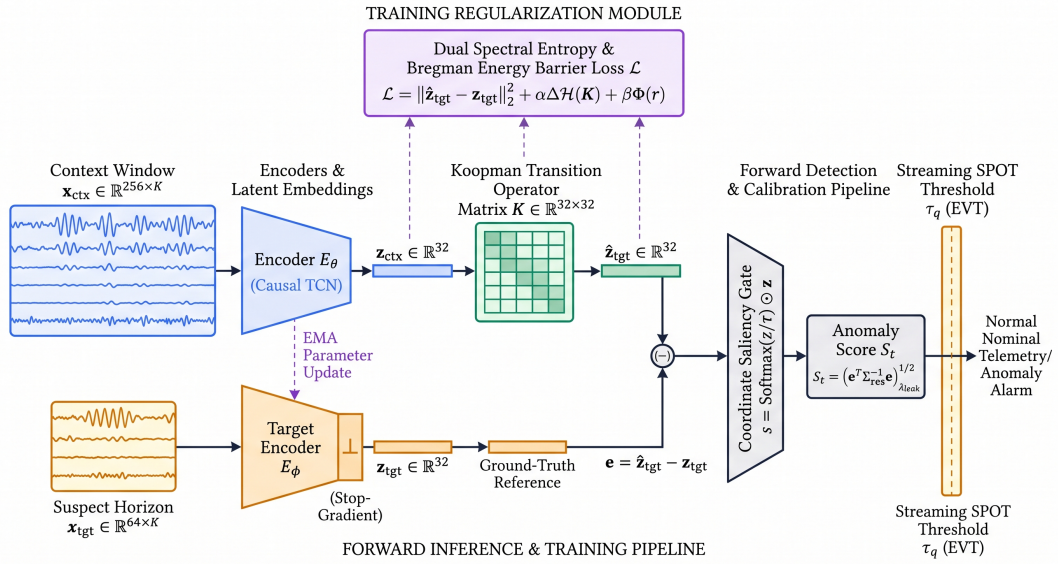


Fig. 1. Operator-Entropy JEPA architecture. Online (E_θ) and momentum target (E_ϕ) encoders map context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows into latent space, where a Koopman operator ($\mathbf{K} \in \mathbb{R}^{32\times32}$) forecasts $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$ under spectral-entropy and Bregman-barrier regularization for coordinate-gated SPOT scoring.

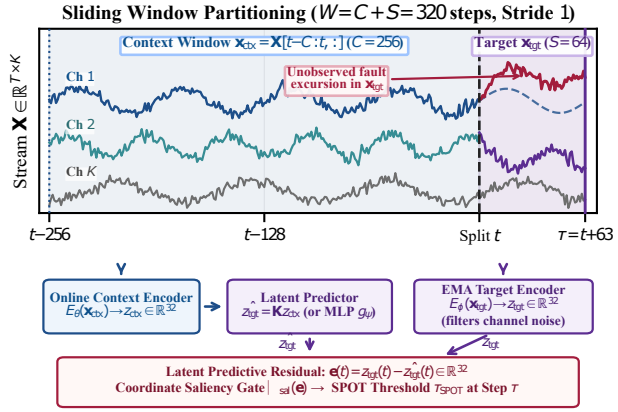


Fig. 2. Sliding window partitioning of a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T\times K}$ into a historical context window $\mathbf{x}_{\text{ctx}}$ (C=256 steps) and a suspect target horizon $\mathbf{x}_{\text{tgt}}$ (S=64 steps, evaluated at decision timestamp $\tau = t + S - 1$). Rather than reconstructing noisy raw waveforms, the online encoder E_θ and predictor forecast the target representation $\hat{z}_{\text{tgt}}$ in a 32-dimensional latent space to compare against the EMA target embedding $z_{\text{tgt}} = E_\phi(\mathbf{x}_{\text{tgt}})$.

(EMA) of the online weights θ, $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with a cosine momentum schedule $m_k \in [0.996, 0.9995]$ [13], [15]. By predicting the summary embedding z_{tgt} from z_{ctx}, the model discards uninformative channel-level jitter while preserving structural trajectory departures.

B. Latent Predictor Architectures and Dynamical Inductive Priors

We formulate Operator-Entropy JEPA as our primary transition architecture and evaluate it alongside an unconstrained multi-layer perceptron predictor (`TS-JEPA`) and two comparative physics-inspired variants (Reynolds-Stress and Potential-Flow):

1) Operator-Entropy JEPA (Spectral Dynamics and Modal Dispersion): Motivated by Koopman operator theory [25], [26], which represents nonlinear dynamical evolution through linear operators acting on learned observable embeddings, we parameterize the latent transition via a learnable linear matrix $\mathbf{K} \in \mathbb{R}^{D\times D}$ such that $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. In unconstrained linear forecasting, gradient descent can allow $\mathbf{K}$ to either concentrate all energy onto a few dominant directions (dimensional rank collapse) or shrink its overall magnitude toward zero (norm collapse). We prevent both failure modes by regularizing the singular value spectrum of $\mathbf{K}$ while separating modal dispersion from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, where each squared singular value $\lambda_i = \sigma_i^2$ represents the amplification energy along the i-th orthogonal mode and $\|\mathbf{K}\|_F^2 = \sum_{i=1}^D \lambda_i$ is the total Frobenius energy. Normalizing the modal energies into a probability distribution $p_i = (\lambda_i + \epsilon_{\log})/\sum_{j=1}^D(\lambda_j + \epsilon_{\log})$ (with $\epsilon_{\log} = 10^{-12}$ for numerical stability) defines the Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^D p_i \log p_i$. We define the *Spectral Entropy Deficiency* as the Kullback–Leibler (KL) divergence between the modal distribution $\mathbf{p}$ and the ideal uniform equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \quad (1)$$

When all D singular modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), $\Delta\mathcal{H}(\mathbf{K}) = 0$; conversely, if $\mathbf{K}$ collapses onto a single rank-1 direction, $\Delta\mathcal{H}(\mathbf{K})$ rises toward $\log D$. Minimizing $\Delta\mathcal{H}(\mathbf{K})$ thus encourages a well-conditioned, near-isometric transition matrix.

To prevent the total energy of $\mathbf{K}$ from shrinking toward zero without resorting to hard gradient clipping, we apply a smooth logarithmic Bregman divergence barrier to the normalized energy ratio $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/D$:

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K}) + \epsilon\big) + \log(1+\epsilon) \geq 0, \quad (2)$$

with $\epsilon = 10^{-6}$. This convex barrier equals zero at the unit-energy scale $r(\mathbf{K}) = 1$ (where $\mathbf{K}$ is initialized near the identity matrix) and smoothly penalizes both energy decay ($r \to 0$) and runaway amplification ($r \gg 1$). Combining both

terms gives the spectral regularizer $\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K})$, yielding the training loss:

$$\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_{b=1}^{B}\left\|\mathbf{K}z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\right\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}, \tag{3}$$

where B is the batch size, $\beta_{\text{norm}} = 1.0$, $\lambda_{\text{ent}} = 0.5$, and $\beta_{\text{Lyap}} = 0.02$. Here $\mathcal{L}_{\text{Lyap}}$ is a discrete Lyapunov covariance-alignment penalty that matches the trace-normalized batch covariance of predicted states $\mathbf{K}\,\text{sg}(z_{\text{ctx}})$ to that of z_{tgt}, applying a stop-gradient operator $\text{sg}(\cdot)$ to z_{ctx} so that the operator regularizers condition $\mathbf{K}$ without distorting the encoder representations.

Proposition 1 (Operator Non-Triviality Under Finite-ϵ and Singular Barriers)**.** *Consider* $\mathcal{L}_\epsilon(\mathbf{K}) = \frac{1}{BD}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent}, \beta_{norm} > 0$ *and empirical context–target cross-covariance* $\mathbf{M} = \frac{1}{BD}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top}$*: (i) Under the finite-*ϵ *smooth barrier (*$\epsilon = 10^{-6}$*), the regularizer gradient vanishes at the origin,* $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$*, because* $r(\mathbf{K})$ *is quadratic in* $\mathbf{K}$*; however, whenever context and target embeddings exhibit non-zero cross-covariance (*$\mathbf{M} \neq \mathbf{0}$*),* $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$*, which excludes* $\mathbf{K} = \mathbf{0}$ *as a stationary point or minimizer. (ii) In the singular-barrier limit (*$\epsilon \to 0^+$*),* $\lim_{\|\mathbf{K}\|_F \to 0}\Phi_{energy}(\mathbf{K}) = +\infty$ *unconditionally.*

Proof. For $\epsilon > 0$, $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{K}) = \frac{2}{D}\Phi'_{\text{energy}}(r)\,\mathbf{K} = \mathbf{0}$ at $\mathbf{K} = \mathbf{0}$, and $\mathbf{K} = \mathbf{0}$ yields uniform smoothed probabilities $p_i = 1/D$, minimizing $\Delta\mathcal{H} \geq 0$ so $\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$. Differentiating the prediction term gives $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$ whenever $\mathbf{M} \neq \mathbf{0}$. As $\epsilon \to 0^+$, $-\log r(\mathbf{K}) \to +\infty$ as $\|\mathbf{K}\|_F \to 0$. □

Remark (Scope of Regularization): Proposition 1 guarantees non-degeneracy of the transition matrix $\mathbf{K}$, not diversity of the encoder representations $z_{\text{ctx}}, z_{\text{tgt}}$. Unlike representation regularizers such as VICReg [16] or Barlow Twins [17] that penalize encoder covariance collapse directly, $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ constrains only the transition spectrum; encoder collapse is prevented empirically by the momentum target encoder and stop-gradient decoupling (Table I(c)).

2) Comparative Physical Priors (Reynolds-Stress and Potential-Flow Heads): To test whether alternative continuum-mechanics priors improve anomaly detection, we evaluate two additional latent predictor heads under identical encoder backbones:

- **Reynolds-Stress JEPA (Second-Moment Closure Head):** Inspired by fluid-mechanical Reynolds decomposition (which separates mean flow from turbulent covariance fluctuations), this variant augments the mean prediction $\hat{z}_{\text{tgt}}$ with a lower-triangular Cholesky factor head $\mathbf{L} \in \mathbb{R}^{d_s \times d_s}$ ($d_s = 16$) that predicts the projected outer-product second moment $\mathbf{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I}$ via $\mathbf{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I}$. Training minimizes the sum of the latent prediction MSE and $0.1\,\text{MSE}(\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}})$, and inference adds $0.5\,g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$ to the primary residual score.
- **Potential-Flow JEPA (Curl-Free Flow Matching Head):** Motivated by conservative irrotational vector fields, this variant parameterizes a curl-free velocity field $\hat{v} = -\nabla_{z_t}\Phi_\psi$ as the negative gradient of a scalar potential Φ_ψ conditioned on flow time $t \in [0, 1]$, context z_{ctx}, and a soft router over $K_r = 4$ orthonormal subspaces. Training uses conditional flow matching along linear paths from Gaussian noise to z_{tgt}, and inference evaluates the velocity residual at the trajectory midpoint. Because curl-free gradient fields cannot represent closed rotational orbits without multi-regime switching, this inductive bias is hypothesized to penalize periodic nominal telemetry (Section VII-B).

C. Anomaly Scoring and Operational Detection Latency

At test time, sliding windows are processed at stride 1. Let $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t) \in \mathbb{R}^D$ denote the latent prediction residual, and let $g(\mathbf{e})$ denote the coordinate saliency gate defined in (5). Unconstrained `TS-JEPA` scores windows directly via $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because the final LayerNorm layer projects latent vectors onto the sphere of radius $\sqrt{D}$ and strips radial amplitude, Operator-Entropy JEPA records the pre-LayerNorm radial energy scaled by window RMS for both encoders ($e_{\text{ctx}}, e_{\text{tgt}}$), forms the predicted radial energy $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, and combines directional and radial discrepancies:

$$\mathcal{S}_{\text{OpEnt}}(t) = g\big(\mathbf{e}(t)\big) + 0.2\,\frac{|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|}{e_{\text{tgt}}(t) + e_{\text{pred}}(t) + 10^{-6}}. \tag{4}$$

Three-Part Operational Causality Framework: To avoid confounding data access, window buffering, and alarm timing, we formalize online evaluation under three criteria (illustrated in Fig. 3):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision timestamp $\tau = t + S - 1$, the detector observes only historical data $\mathbf{X}[t - C : \tau, :]$; neither encoder accesses observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):** Because z_{tgt} summarizes the S-step target block $\mathbf{X}[t : t + S, :]$, window score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 3, Row 3). This creates a structural score-emission latency of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This structural lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$, which measures the elapsed steps from ground-truth anomaly onset t^* to the first timestamp $t_{\text{alarm}} \geq t^*$ whose attributed score exceeds threshold τ_{SPOT}. Shortening the horizon to $S \in \{1, 4\}$ eliminates score-emission latency when rapid tripwire response is required (Section VII-D).

(c) **Strictly causal trailing vs. retrospective buffered attribution:** Under *strictly causal trailing attribution*, window score s_t is assigned only to the decision timestamp $\tau = t + S - 1$ for zero-lookahead streaming. Under *retrospective buffered smear attribution*, s_t is averaged across all steps in $[t + 1, t + S]$, which introduces fixed-lag smoothing and can smear post-onset scores up to S steps before t^*. Both regimes are reported separately throughout (Table II(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating: In high-dimensional streams ($K \geq 25$), faults affecting a single subsystem can be diluted by nominal variance across the remaining coordinates. To amplify sparse latent deviations without losing global Euclidean norm information, we apply a softmax coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ to the residual vector $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \qquad w_d = \frac{\exp(|e_d|/\tau_g)}{\sum_{j=1}^{D}\exp(|e_j|/\tau_g)}, \tag{5}$$

with fixed blend $\alpha = 0.5$ and temperature $\tau_g = 0.5$ across all latent models. Input channels are encoded with uniform weights in the deployed pipeline; an auxiliary context-variance input-channel gate is evaluated in Section VI-C.

2) Multiscale Frequency Front-End: Before the dilated TCN backbone, each input channel is split across $J = 4$ smooth dyadic frequency shells on a 256-point real Fast Fourier Transform (rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J}\Psi_j[\omega_m] \equiv 1$ to within 1.2×10^{-7}). Each shell is processed by a causal 1D convolution with kernel sizes $\{15, 9, 5, 3\}$ from coarse to fine scales and recombined via learned softmax routing. Because smooth Fourier shells overlap at band edges, this module acts as a multiscale frequency front-end rather than a strictly orthogonal wavelet bank (Table I(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert continuous scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ using nominal calibration data alone, we apply Streaming Peaks-Over-Threshold (SPOT) [7]. Rather than assuming the entire score distribution is Gaussian, SPOT isolates the upper tail above the 98th calibration percentile $u = p_{98}$ and fits the exceedances $Y = \mathcal{S}(t) - u > 0$ to a Generalized Pareto Distribution (GPD) with tail-heaviness shape parameter ξ and scale parameter σ via Grimshaw's maximum-likelihood algorithm [38]. Inverting the fitted GPD tail at target false-alarm probability $q = 10^{-3}$ (one expected false alarm per 1,000 nominal steps) yields the closed-form threshold:

$$\tau_{\text{SPOT}} = u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], \tag{6}$$

where N_{cal} is the number of nominal calibration windows and N_u is the number of exceedances above u. All primary benchmarks use standard unadjusted SPOT (τ_{SPOT}). In Section VII-F, we also audit two heuristic kurtosis adjustments: total-threshold scaling $\tau_{\text{eff}} = \tau_{\text{SPOT}}[1+\frac{1}{6}\tanh(\gamma_2)]$ and excess-only scaling $\tau_{\text{exc}} = u + (\tau_{\text{SPOT}} - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$.

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures are trained and evaluated under an identical streaming time-series pipeline using a past context window of $C = 256$ steps and a suspect target horizon of $S = 64$ steps (total window length $W = C + S = 320$), evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (48 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, and channel statistics) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW (initial learning rate 10^{-3} decayed to 10^{-5} via cosine annealing, weight decay 10^{-4}, batch size 32).

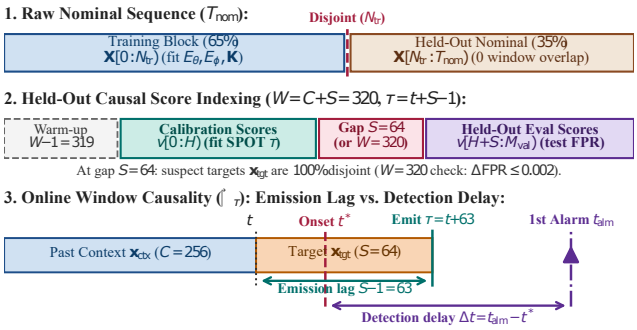


Fig. 3. Chronological split, boundary window indexing, and online causality protocol (C=256, S=64, W=320). Row 1 separates raw training (65%) and held-out nominal (35%) observations with zero window overlap across N_{tr}. Row 2 indexes causal held-out scores $v[0 : M_{\text{val}})$ into context warm-up ($W-1$=319), calibration $v[0 : H)$, isolation gap S=64 (or full-window W=320), and held-out evaluation $v[H+S : M_{\text{val}})$. Row 3 distinguishes structural score-emission lag ($S-1$=63) from empirical fault-detection delay ($\Delta t = t_{\text{alm}} - t^*$).

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}}\times K}$ is partitioned chronologically (Fig. 3, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 3, Row 2). Under causal trailing attribution, the first $W-1 = 319$ observations serve as context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint; the first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A stricter full-window isolation check ($G = W = 320$, excluding $v[H : H + 320)$ so even passive context is disjoint across the seven streams with $M_{\text{val}}/2 > 420$) leaves stream-macro causal FPR virtually unchanged (2.07% $\rightarrow$ 2.18% for OpEntropy and 3.02% $\rightarrow$ 3.24% for TS-JEPA versus 12.15% $\rightarrow$ 13.97% for TimesNet and 9.05% $\rightarrow$ 9.73% for TranAD). All primary tables use fixed standard SPOT thresholds during testing.

b) Multi-Seed Replication, Non-Parametric Hypothesis Testing, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds. Because random seeds on the same stream are not independent observations, all metrics are averaged across seeds within each stream first, treating the telemetry stream ($n = 33$ or $n = 19$) as the independent statistical unit. To compare models without assuming Gaussian error distributions across diverse domains, we employ three statistical tools:

1) **Paired Wilcoxon signed-rank test and FDR adjustment:** The Wilcoxon signed-rank test is a non-parametric paired hypothesis test that evaluates whether the median difference between two models across the same streams is zero by ranking the absolute magnitudes of their per-stream score differences. Unlike a paired Student's t-test, it prevents a single extreme outlier stream from skewing the comparison. When testing multiple model pairs simultaneously, we adjust p-values using the Benjamini–Hochberg false-discovery-rate (FDR) procedure ($q_{\text{FDR}} = 0.05$) to cap the expected proportion of false statistical positives, and report 95% stream-resampled bootstrap confidence intervals (2,000 resamples).
2) **Two One-Sided Tests (TOST) for statistical equivalence:** In a standard two-sided test, an insignificant result ($p > 0.05$) indicates only inconclusive evidence rather than equal performance. To test whether a physics-inspired regularizer produces practically negligible change relative to an unregularized control, TOST reverses the null hypothesis by testing whether the true mean difference $\mu_1 - \mu_2$ lies outside a prespecified equivalence margin $[-\Delta, +\Delta]$ via two one-sided tests: $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}). Formal equivalence is established if $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ (meaning the 90% paired confidence interval falls strictly inside $[-\Delta, +\Delta]$). We prespecify $\Delta = \pm 0.03$ Point-F1 for fixed-backbone regularizer controls ($\approx 13\%$ of baseline F1, $< 0.09\times$ between-stream standard deviation) and $\Delta = \pm 0.05$ for cross-architecture comparisons, and report narrow-margin sensitivity checks down to $\Delta \in \{\pm 0.010, \pm 0.015, \pm 0.020\}$ (Table I(a)).
3) **Distributional tail and drift diagnostics (γ_2 and D_{KS}):** To quantify why anomaly scores trigger false alarms, we measure two summary statistics on nominal data: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$, which measures tail heaviness relative to a Gaussian distribution ($\gamma_2 = 0$), where *leptokurtic* scores ($\gamma_2 > 0$) exhibit heavy outlier-prone tails and *platykurtic* scores ($\gamma_2 < 0$) exhibit light, flat-topped tails compressed by normalization or temporal smoothing; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0, 1]$, which measures the maximum vertical gap between the empirical cumulative distribution functions of the calibration block (F_{cal}) and the held-out nominal evaluation block (F_{hold}), directly quantifying how severely the nominal baseline shifts over time.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

We separate primary multi-seed conclusions (Cohorts 1–2 across 12 neural, linear, and non-parametric detectors) from secondary single-seed cross-checks. Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation:

1) **Native published window lengths** (5 streams): Evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their published window sizes raises Point-F1 (0.164 and 0.218), but observation residuals remain severely leptokurtic ($\gamma_2 = 448.3$ and 98.1 vs. 3.1 for TS-JEPA) with high nominal FPR (28.4% and 44.1%).
2) **Secondary 130-stream single-seed cross-check:** Reconstructors replicate their published point-adjusted F1 on SMD and MSL, and Table II(d) provides a secondary 130-stream, 11-dataset single-seed comparison against NCAD, DCdetector, and Anomaly Transformer.

C. Cohort Summary and Reproducibility Specifications

Experiments are organized into five non-overlapping cohorts:

- **Cohort 1 (33-stream primary multi-seed benchmark):** 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), reported in Tables II(a)–(c) and IV(a).
- **Cohort 2 (19-stream mechanism and operating-curve suite):** 12 controlled backbone arms across 5 seeds, reported in Table I(a),(e), Table III(b), and Fig. 4.
- **Cohort 3 (6-stream operator sweep and quiet-fault suite):** Factorial regularizer sweep (3 seeds) and six injected single-channel fault types, reported in Table I(c),(d).
- **Cohort 4 (5-stream horizon and frequency-shell panels):** Target-horizon latency sweep ($S \in \{1, 4, 16, 32, 64\}$, 3 seeds) and dyadic front-end ablation, reported in Table III(a) and Table I(b).
- **Cohort 5 (130-stream single-seed baseline screen):** Secondary cross-check across 11 datasets against NCAD, DCdetector, and Anomaly Transformer (Table II(d)).

At $K = 25$ channels, model parameter counts are comparable across families: 285,701 (OpEntropy JEPA), 289,061 (TS-JEPA), 296,037 (Reynolds-Stress), 338,209 (Potential-Flow), 648,553 (TimesNet), and 205,746 (TranAD). Non-finite telemetry entries are zero-imputed with channel standard deviations floored at 10^{-2}; SPOT fits GPD parameters (ξ, σ) above $u = p_{98}$ via Grimshaw MLE with L-BFGS-B fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$). All 195 primary multi-seed runs converged under the standard 20-epoch schedule on PyTorch 2.4 / CUDA 12.4 (NVIDIA RTX 5090, 0.60–0.84 ms per window, 184 MB VRAM); stream manifests, fitted thresholds, and source code are available at https://github.com/quaid281/NCAD.

D. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [10], [31], we evaluate

models across three complementary tiers: (i) *primary calibration and accuracy endpoints*, comprising held-out nominal FPR, unadjusted Point-F1, and threshold-free PR-AUC; (ii) *operating characteristics*, evaluating event recall across deployable calibration quantiles (q^*, Table III(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in [0.01, 0.50]$, Fig. 4(b)); and (iii) *secondary operational endpoints*, reporting unweighted Dataset-Macro Point-F1 (preventing the 14-stream GHL block from dominating), event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\mathrm{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at anomaly prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether physics-inspired regularizers drive anomaly detection accuracy, Table I(a) disables each physical penalty on its fixed architecture across all 19 non-GHL streams:

1) **Operator-Entropy vs. λ_{ent}=0 (fixed Koopman backbone):** Removing the spectral regularizer leaves buffered Point-F1 (0.232 vs. 0.235, paired difference -0.004, two-sided $p = 0.89$) and PR-AUC (0.365 vs. 0.365) virtually identical. Two One-Sided Tests confirm formal equivalence within the prespecified $\Delta = \pm 0.03$ margin ($p_{\text{TOST}} = 0.0007$) and across narrow-margin sensitivity checks down to $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 0.0015$), $\Delta = \pm 0.015$ ($p_{\text{TOST}} = 0.0024$, with 90% CI $[-0.0112, +0.0029]$ strictly inside $[-\Delta, +\Delta]$), and $\Delta = \pm 0.010$ ($p_{\text{TOST}} = 0.0088$), as well as within ± 0.010 PR-AUC and ± 0.025 event recall.
2) **Reynolds-Stress vs. α_{stress}=0 (fixed backbone):** Disabling the second-moment loss yields a $+0.013$ Point-F1 shift (two-sided $p = 0.59$), confirming TOST equivalence within $\Delta = \pm 0.05$ ($p_{\text{TOST}} = 0.0019$) and down to $\Delta = \pm 0.035$ ($p_{\text{TOST}} = 0.0031$).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling multi-regime routing shifts Point-F1 by -0.054 ($p_{\text{TOST}} = 0.17$), failing equivalence at $\Delta = \pm 0.05$.

These fixed-backbone regularizer controls must be distinguished from cross-architecture comparisons against unconstrained TS-JEPA (which uses an MLP predictor without radial energy scoring or Lyapunov covariance alignment). On the 19-stream subset, the $+0.028$ Point-F1 difference between OpEntropy JEPA and TS-JEPA (90% CI $[-0.018, +0.074]$, $p_{\text{TOST}} = 0.11$) is statistically inconclusive at $\Delta = \pm 0.05$, whereas across all 33 primary streams OpEntropy JEPA ($+0.016$, 90% CI $[-0.009, +0.045]$, $p_{\text{TOST}} = 0.0019$) and Reynolds-Stress (-0.008, $p_{\text{TOST}} = 0.0017$) satisfy TOST(± 0.05) (with OpEntropy vs. TS-JEPA also satisfying TOST(± 0.030) at $p_{\text{TOST}} = 0.0077$ and ± 0.015 causal FPR at $p_{\text{TOST}} = 0.0001$).

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

Table I(c) crosses the spectral-entropy term and Bregman energy barrier across $\lambda_{\text{ent}} \in [0, 2.0]$ on six streams (three seeds). To quantify spectral diversity, we report the effective rank $\mathrm{RankMe} = \exp(-\sum_i p_i \log p_i) \in [1, 32]$ (which measures how many of the $D = 32$ dimensions carry active variance) and the operator condition number $\kappa = \sigma_{\max}/\sigma_{\min} \geq 1$ (the ratio of largest to smallest singular values, where $\kappa = 1$ denotes an ideal distance-preserving isometry). Three findings emerge. First, because the Koopman matrix $\mathbf{K}$ is initialized near the identity, the unregularized operator is already near full rank ($\mathrm{RankMe}(\mathbf{K}) = 31.53$ of 32, condition number $\kappa = 1.437$); increasing λ_{ent} tightens conditioning monotonically ($\kappa = 1.255$ at the deployed $\lambda_{\text{ent}} = 0.50$, and $\kappa = 1.154$ at $\lambda_{\text{ent}} = 2.00$) while lowering training loss variation ($\mathrm{CV}_{\mathcal{L}} : 0.565 \rightarrow 0.545$). Second, because $\mathcal{L}_{\text{Lyap}}$ applies a stop-gradient to z_{ctx}, encoder effective ranks remain invariant across all regularizer strengths ($\mathrm{RankMe}(z_{\text{ctx}}) \in [5.16, 5.54]$, $\mathrm{RankMe}(z_{\text{tgt}}) \in [11.54, 11.58]$). Third, detection and false-alarm rates remain stable across arms, with the deployed setting ($\lambda_{\text{ent}} = 0.50, \beta_{\text{norm}} = 1$) attaining the highest causal event recall (0.444 vs. 0.369 unregularized). On an eight-stream decoupling panel, pairing detached Lyapunov alignment ($\beta_{\text{Lyap}} = 0.02$) with the smooth Bregman barrier cuts causal detection delay from 127.0 steps (TS-JEPA) and 80.2 steps ($\beta_{\text{Lyap}} = 0$) to 58.2 steps while raising causal Point-F1 from 0.044 to 0.094.

C. Ablation 3: Auxiliary Input-Channel Gating and Quiet-Fault Monitors

Comparing the auxiliary context-variance input-channel gate against uniform channel weighting on four streams confirms that uniform weighting achieves higher macro Point-F1 (0.118 vs. 0.077), so the deployed pipeline retains uniform input encoding. In Table I(d), injecting six single-channel fault types into six multivariate streams shows that quiet faults (flatlines, stuck sensors, dropouts) yield low point recall (0.02–0.07) across all latent gating variants because LayerNorm embeddings absorb constant single-channel offsets as nominal background. Auxiliary observation-space monitors, namely a rolling variance floor (`lowvar`) and maximum channel z-score (`dev_maxz`), recover quiet-fault recall (0.18–0.32) at the cost of inflating nominal FPR eight-fold (14.1%–14.9% vs. 1.9%).

D. Ablation 4: Multiscale Dyadic-Shell Front-End

In Table I(b), replacing the standard convolutional stem with the 4-shell Fourier front-end increases macro PR-AUC from 0.213 to 0.263 ($+0.050$, led by SMD and GECCO) and lowers calibration residual kurtosis ($1.409 \rightarrow 1.069$), while macro Point-F1 shifts from 0.118 to 0.078 due to conservative threshold tightening on SMAP `P-3`. The module thus acts as a ranking-conditioning stem rather than a uniform F1 gain.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 12 controlled backbone arms across all 19 non-GHL streams in Table I(e) and Fig. 4(a) separates geometric score compression from representation drift reduction:

1) **Unnormalized observation forecasting (`TCN-ObsPred`):** Raw observation residuals exhibit

TABLE I
ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE REGULARIZER CONTROLS AND TOST EQUIVALENCE (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), (C) OPERATOR REGULARIZER SWEEP (n=6, 3 SEEDS), (D) INJECTED QUIET-FAULT POINT RECALL (n=6), AND (E) 12-ARM BACKBONE COMPARISON (n=19, 5 SEEDS).

(a) Regularizer Controls on Fixed Architectures & TOST Equivalence

Model / Control	Point-F1	PR-AUC	FPR	ER	Paired ΔF1 (90% CI)	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline (cross-arch)	–
Operator-Entropy JEPA	0.233 ± 0.028	0.351	0.101	0.550	-0.004 $[-0.011, +0.003]$	0.0007 ($\pm.03$)
$\lambda_{\text{ent}} = 0$ control	0.235 ± 0.038	0.365	0.099	0.580	(also $\pm.015$: p=0.0024)	0.0088 ($\pm.01$)
Reynolds-Stress JEPA	0.190 ± 0.022	0.313	0.074	0.457	$+0.013$ $[-0.001, +0.030]$	0.0019 ($\pm.05$)
$\alpha_{\text{stress}} = 0$ control	0.182 ± 0.036	0.299	0.067	0.472	(also $\pm.035$: p=0.0031)	–
Potential-Flow JEPA	0.138 ± 0.044	0.235	0.087	0.449	-0.054 $[-0.131, +0.012]$	0.17 (fails)
no-regime control ($K_r = 1$)	0.189 ± 0.024	0.279	0.084	0.431	–	–
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583	–	–
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490	–	–

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR-AUC)	Standard Conv	Dyadic Shells	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	-0.186 / $+0.017$
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	$+0.002$ / -0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	$+0.068$ / $+0.193$
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	-0.048 / -0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	-0.035 / $+0.121$
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	-0.040 / $+0.050$

(c) Factorial and Strength Sweep of the Operator Regularizer

			Operator / representation / stability				Buffered			Causal		
Arm	λ_{ent}	β_{norm}	RankMe($\mathbf{K}$)	$\kappa(\mathbf{K})$	RankMe($z_{\text{ctx}}/z_{\text{tgt}}$)	$\text{CV}_{\mathcal{L}}$	FPR	F1	ER	FPR	F1	ER
Neither	0	0	31.53	1.437	5.16/11.57	0.565	0.071	0.044	0.250	0.017	0.009	0.369
Entropy only	0.5	0	31.84	1.256	5.25/11.57	0.550	0.071	0.040	0.266	0.017	0.011	0.389
Barrier only	0	1	31.54	1.431	5.37/11.58	0.563	0.074	0.047	0.266	0.016	0.011	0.389
Both	0.05	1	31.61	1.395	5.16/11.57	0.562	0.074	0.039	0.250	0.017	0.008	0.373
Both	0.10	1	31.67	1.363	5.19/11.58	0.559	0.073	0.039	0.250	0.017	0.008	0.421
Both (deployed)	0.50	1	31.85	1.255	5.54/11.56	0.549	0.072	0.038	0.266	0.017	0.010	0.444
Both	1.00	1	31.91	1.204	5.25/11.55	0.546	0.070	0.037	0.250	0.017	0.007	0.365
Both	2.00	1	31.95	1.154	5.34/11.54	0.545	0.072	0.032	0.250	0.017	0.004	0.397

(d) Quiet-Fault Point Recall

Injected fault	Unif.	Var.	MAD	`lowvar`	`dev_maxz`
Flatline	0.021	0.043	0.058	0.182	0.087
Flatline (mid-var.)	0.024	0.048	0.068	0.317	0.087
Stuck sensor	0.021	0.043	0.058	0.182	0.087
Dropout	0.020	0.043	0.058	0.182	0.087
Low-amp. step	0.030	0.049	0.062	0.182	0.087
$+5\sigma$ spike	0.023	0.042	0.059	0.182	0.108
Mean nom. FPR	0.019	0.069	0.055	0.149	0.141

(e) Controlled Mechanism-Discrimination Study (19 Non-GHL Streams)

	Tail/Drift		Held FPR		Point-F1		PR-AUC		Causal Event Recall			
Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA`	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`OpEntropy`	0.94	0.209	0.028	0.101	0.153	0.233	0.280	**0.351**	0.517	0.752	0.371	0.573
`JEPA-CtxRecon`	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	**0.431**	0.573
`JEPA-NoEMA`	1.16	**0.167**	**0.009**	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
`TS-JEPA-NoGate`	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN`	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	**0.624**
`TCN-ObsPred-LN`	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest`	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred`	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon`	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet`	5.54	0.438	0.243	0.300	**0.209**	0.197	**0.293**	0.329	**0.588**	0.618	0.217	0.407
`TranAD`	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

heavy tail kurtosis ($\gamma_2 = 6.10$), severe calibration-to-holdout KS drift ($D_{\text{KS}} = 0.463$), and high causal false alarms (24.7%).

2) **LayerNorm observation bottleneck (`TCN-ObsPred-LN`):** Inserting a $D = 32$ LayerNorm bottleneck before decoding mechanically bounds excursions, compressing kurtosis to $\gamma_2 = 0.91$ and halving causal FPR to 11.7%, but leaves calibration-to-holdout drift elevated ($D_{\text{KS}} = 0.376$).
3) **Learned latent projection (`TS-JEPA`, `OpEntropy`):** Forecasting in learned latent coordinates halves calibration-to-holdout KS drift (D_{KS} : $0.376 \rightarrow 0.214$ and 0.209, paired Wilcoxon $p \leq 0.001$) and reduces causal false alarms an additional five- to eight-fold (1.7%–2.8%).
4) **Context autoencoding (`JEPA-CtxRecon`) and linear controls:** Autoencoding context embeddings without temporal foresight achieves comparable drift reduction ($D_{\text{KS}} = 0.203$, causal FPR 1.7%, event recall 0.542), while linear projection (`PCA-LinPred-LN`) reaches 2.1% causal FPR at lower F1 (0.017) and `IsoForest` collapses in recall (0.149). LayerNorm geometry thus bounds tail kurtosis, while learned latent projection is required to suppress temporal distribution drift.

VII. RESULTS AND DISCUSSION

A. Benchmark Performance on Continuous Telemetry

Table II summarizes performance across all 33 continuous telemetry streams under the chronological held-out SPOT calibration protocol ($u = p_{98}, q = 10^{-3}$), reporting retrospective buffered attribution in Panel (a), strictly causal streaming in Panel (b), and per-dataset breakdowns in Panel (c). The central empirical finding is substantial relative false-alarm suppression under automated thresholding, whereas overall Point-F1 and PR-AUC differences against reconstructive baselines are statistically inconclusive.

Relative False-Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams (Table II(a),(b)), latent predictive scoring via TS-JEPA and OpEntropy JEPA lowers held-out nominal FPR three- to four-fold under buffered attribution (9.3% and 10.1% versus 30.0% for TimesNet and 30.7% for TranAD) and eight- to twenty-fold under strictly causal streaming (2.4% and 2.8% versus 24.3% and 22.2%). Across all 33 streams, the paired buffered FPR reduction is statistically significant for both OpEntropy JEPA (-14.4% vs. TimesNet, 95% CI $[-24.1\%, -6.5\%]$, $p = 0.006$; -14.2% vs. TranAD, $p = 0.006$) and TS-JEPA (-15.3%, $p = 0.002$; -15.1%, $p = 0.003$). Dataset-clustered permutation tests across the $N = 7$ domain families remain significant ($p = 0.016$–0.031), and leave-one-dataset-out (LODO) folds hold macro buffered FPR to 3.8%–7.0% for JEPAs versus 8.4%–18.3% for recon-

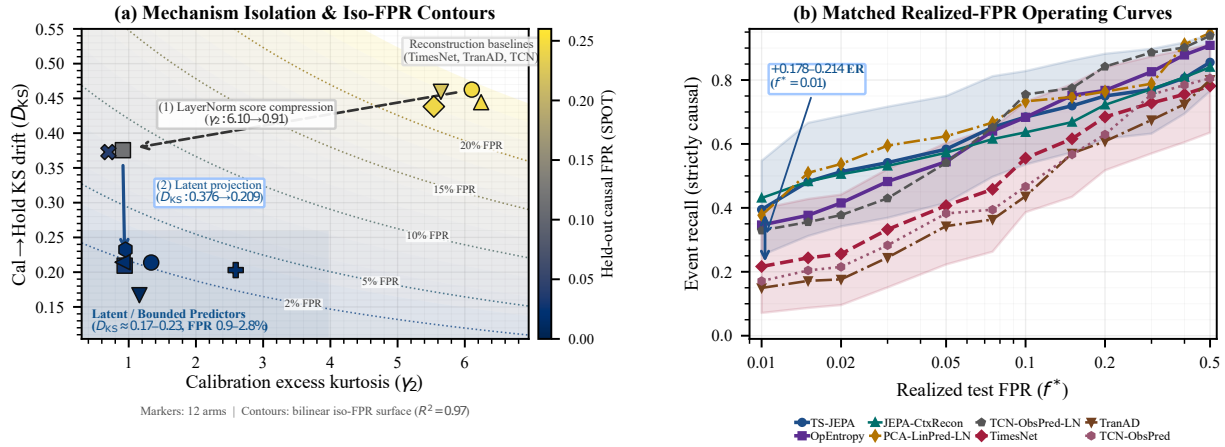


Fig. 4. Residual conditioning and operating curves across 19 non-GHL streams (five seeds, strictly causal trailing inference). **(a)** Calibration excess kurtosis (γ_2) vs. calibration-to-holdout KS drift (D_{KS}), colored by held-out causal FPR at SPOT $q{=}10^{-3}$ with bilinear iso-FPR topographic contours ($R^2{=}0.97$), illustrating LayerNorm score compression and learned latent projection. **(b)** Causal event recall across matched realized test-FPR budgets $f^* \in [0.01, 0.50]$ with 95% stream-bootstrap confidence bands.

TABLE II

HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}, q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL STREAMS (95% STREAM-BOOTSTRAP CIS IN BRACKETS) AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN (MEAN ± SD), AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Buffered, non-GHL (19 streams)					(b) Strictly causal, non-GHL (19 streams)				GHL buf (14)		Summary
Model	FPR [95% CI] ↓	F1 [95% CI] ↑	PR-AUC ↑	ER ↑	Δt ↓	FPR [95% CI] ↓	F1 [95% CI] ↑	ER ↑	Δt ↓	FPR ↓	F1 ↑	DS-Macro F1 ↑
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093 [.048, .145]	0.206 [.113, .319]	0.331	0.477	110.3	0.024 [.007, .046]	0.161 [.063, .287]	0.487	124.0	0.004	0.000	0.115
Operator-Entropy JEPA	0.101 [.043, .180]	**0.233** [.131, .355]	**0.351**	0.550	79.5	0.028 [.007, .063]	0.153 [.070, .247]	0.517	112.5	0.012	0.001	**0.125**
Reynolds-Stress JEPA	0.074 [.038, .118]	0.190 [.101, .295]	0.313	0.457	160.9	0.011 [.006, .017]	0.131 [.051, .232]	0.491	142.1	**0.003**	0.002	0.101
Potential-Flow JEPA	0.087 [.041, .142]	0.138 [.062, .231]	0.235	0.449	104.1	**0.008** [.002, .016]	0.029 [.005, .064]	0.459	138.4	0.011	0.006	0.089
Observation-space reconstructive baselines												
TimesNet	0.300 [.162, .446]	0.197 [.098, .315]	0.329	**0.583**	**48.5**	0.243 [.123, .375]	**0.209** [.109, .326]	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307 [.177, .445]	0.130 [.053, .232]	0.287	0.490	234.4	0.222 [.119, .337]	0.140 [.058, .241]	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005** [.001, .011]	0.051 [.012, .104]	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset descriptors				Buffered Point-F1 (mean ± SD)				Buffered held-out FPR			
Dataset	n	K	π	$F1_{\text{all-pos}}$	TS-JEPA	OpEnt	TimesNet	TranAD	TS-JEPA	OpEnt	TimesNet	TranAD
Daphnet	6	9	.148	.256	0.145±0.058	0.122±0.061	0.229±0.232	0.144±0.119	0.091	0.059	0.313	0.298
GHL	14	16	.013	.026	0.000±0.000	0.001±0.003	0.022±0.030	0.020±0.025	0.004	0.012	0.061	0.046
Genesis	1	18	.003	.006	0.000±0.000	0.008±0.017	0.000±0.000	0.000±0.000	0.017	0.082	0.000	0.000
MSL	4	55	.255	.381	0.161±0.119	0.323±0.263	0.158±0.146	0.151±0.149	0.095	0.177	0.133	0.241
SMAP	6	25	.150	.236	0.379±0.361	0.396±0.329	0.289±0.334	0.167±0.341	0.135	0.125	0.549	0.513
Metro	1	5	.001	.003	0.000±0.000	0.000±0.000	0.001±0.001	0.000±0.000	0.014	0.026	0.000	0.000
SWAN	1	38	.279	.437	0.119±0.048	0.024±0.031	0.000±0.000	0.000±0.000	0.000	0.000	0.000	0.000

(d) 130-stream cross-check (Point-F1, dataset-macro / stream mean): NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA 0.160/0.186.

structors. On MSL, TimesNet achieves a lower buffered FPR than OpEntropy (13.3% vs. 17.7%), while on Genesis, Metro, and SWAN reconstructors emit zero alarms and achieve zero F1 due to overly high calibration thresholds.

Point-F1, Event Recall, and Ranking Parity: Improved false-alarm control does not translate into universal detection superiority. Across the 19 non-GHL streams in Table II(a), buffered Point-F1 and event recall for TS-JEPA (0.206, recall 0.477) and OpEntropy JEPA (0.233, recall 0.550) are comparable to TimesNet (0.197, recall 0.583) and TranAD (0.130, recall 0.490). Across all 33 streams, paired differences against TimesNet (+0.012 Point-F1, $p = 0.54$; −0.000 PR-AUC, $p = 0.39$) and TranAD (+0.051 Point-F1, $p = 0.88$; +0.023 PR-AUC, $p = 0.16$) are statistically indistinguishable from zero. As shown in Table II(c), between-stream standard deviations within each dataset are comparable to the means; only SMAP exceeds the trivial constant-alarm baseline $F1_{\text{all-pos}} = 0.236$ (0.379 and 0.396), although threshold-free PR-AUC exceeds base prevalence π on both SMAP (0.504–0.525 vs. $\pi = 0.150$) and MSL (0.422–0.434 vs. $\pi = 0.255$).

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained regime:* Low-rate telemetry (0.1 Hz spacecraft or hourly Metro traffic) where a 63-step score-emission lag at $S = 64$ (10.5 min to 63 h) is prohibitive for immediate tripwires and requires short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Flatlines, stuck sensors, and dropouts (0.02–0.07 point recall, Table I(d)), which LayerNorm embeddings absorb as nominal background and which require auxiliary variance monitors at higher alarm cost (14%–15% FPR).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Evaluating Physical Inductive Priors: Conditioning vs. Detection

What does the Operator-Entropy formulation contribute beyond unconstrained TS-JEPA? Isolating the spectral regularizer on the fixed Koopman backbone ($\lambda_{\text{ent}} = 0$ control, Table I(a)) confirms strict TOST equivalence down to $\Delta = \pm 0.015$ and ± 0.010 Point-F1 (Section VI-A). However, the OpEntropy transition architecture (combining the linear Koopman operator $\mathbf{K}$, detached Lyapunov covariance alignment, and pre-LayerNorm relative radial energy scoring) shortens mean buffered detection delay from 110.3 steps in TS-JEPA to 79.5 steps (and from 124.0 to 112.5 steps causally in Table II(b), or $127.0 \rightarrow 58.2$ steps on the eight-stream decoupling panel). Cross-architecture Point-F1 differences against TS-JEPA satisfy TOST(± 0.05) across all 33 streams for both OpEntropy ($+0.016$, $p_{\text{TOST}} = 0.0019$) and Reynolds-Stress (-0.008, $p_{\text{TOST}} = 0.0017$), whereas Potential-Flow degrades Point-F1 (-0.081, $p = 0.033$ unadjusted) because its curl-free gradient parameterization penalizes periodic nominal orbits. The spectral-entropy and Reynolds priors therefore act as well-posed conditioning constraints rather than detection drivers.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Strictly causal trailing attribution represents real-time streaming deployment, whereas buffered attribution represents retrospective fixed-lag analysis. Comparing Panels (a) and (b) of Table II highlights a clear operational trade-off. Eliminating the 64-step lookahead buffer lowers nominal false alarms from 9.3%–10.1% to 2.4%–2.8% for TS-JEPA and OpEntropy JEPA (and 1.1% for Reynolds-Stress), compared to 22.2%–24.3% for TranAD and TimesNet. Without retrospective smoothing, causal Point-F1 (0.153–0.161 vs. 0.209 for TimesNet) and causal event recall (0.487–0.517 vs. 0.588) are more conservative, and mean onset-to-alarm delay increases ($\Delta t = 112.5$–124.0 steps vs. 62.2 steps for TimesNet). In online streaming without lookahead buffers, latent predictive scoring thus functions as a conservative high-precision filter that trades raw onset speed for an 8–20$\times$ reduction in false alarms. On five non-stationary incident streams, pairing latent scores with dynamic rolling SPOT suppresses false alarms by up to 21$\times$ under regime drift.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

At target horizon $S = 64$, window scores emit $S - 1 = 63$ steps after window opening, which is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$ measured after anomaly onset (Fig. 3, Row 3). To evaluate whether shorter horizons eliminate emission latency without sacrificing false-alarm control, Table III(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero emission lag), OpEntropy JEPA holds causal FPR to 0.5% (vs. 2.5% for TimesNet) while achieving 0.729 event recall and a detection delay of $\Delta t = 76.3$ steps. As horizon S increases from 1 to 64, causal FPR for TimesNet rises nearly nine-fold ($2.5\% \rightarrow 21.3\%$), whereas OpEntropy and TS-JEPA remain bounded within 0.3%–2.0%. Time-critical monitoring applications can therefore operate at $S \in \{1, 4\}$ to eliminate score-emission latency while preserving relative false-alarm suppression.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

Because reconstructive baselines calibrate to a much higher realized false-alarm rate (22%–31%) than latent predictors (1%–10%), raw recall at a single SPOT threshold confounds discriminative sensitivity with threshold placement. Indeed, when anomaly segments span hundreds to over a thousand timesteps (mean length 402 steps in Daphnet and 1,102 steps in SMAP), a detector alarming randomly at 30% FPR intersects $> 95\%$ of segments by chance alone ($1 - (1 - \text{FPR})^L$). We therefore evaluate two complementary operating regimes:

1) **Deployable nominal calibration quantiles (q^*, Table III(b)):** Thresholds are set at fixed empirical quantiles of the nominal calibration block without test labels. At causal calibration quantile $q^* = 0.050$, TS-JEPA and OpEntropy JEPA achieve higher event recall (0.730 and 0.752) and higher precision (0.251 and 0.259) at fewer false alarms (177–203 per 1,000 nominal steps) than TimesNet (0.618 recall, 262 alarms) and TranAD (0.602 recall, 270 alarms).
2) **Post-hoc matched realized test-FPR curves (f^*, Fig. 4(b) and Table I(e)):** Oracle thresholds are chosen to equalize realized test nominal FPR across models. At a stringent 1% realized causal false-alarm budget ($f^* = 0.01$), latent models nearly double event recall (0.371–0.431) compared to TimesNet (0.217) and TranAD (0.149). At relaxed budgets ($f^* \geq 0.10$ causal and across buffered mappings), recall curves converge ($p = 0.625$ over $f^* \in [0.10, 0.50]$). Latent scoring thus specifically improves sensitivity in the low-false-alarm operating regime ($f^* \leq 0.01$).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does static extreme-value thresholding achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample? It does not. Across all 33 streams (Table IV(a)), standard SPOT achieves stream-macro (cell-weighted) causal FPR of 1.51% (1.71%) for TS-JEPA and 1.95% (2.12%) for OpEntropy, which is roughly 15–20$\times$ the 0.1% design target, and buffered stream-macro (cell-weighted) FPR of 5.54% (6.40%) and 6.38% (7.25%), or 55–64$\times$ (64–73$\times$) the design target. Although this is an order-of-magnitude improvement over TimesNet (16.11% causal, 19.85% buffered) and TranAD (14.23% causal, 19.63% buffered; 142–222$\times$ overshoot), a 1.5%–6.4% nuisance-alarm rate still falls short of strict 0.1% operational targets. Closing this remaining 15–64$\times$ calibration gap requires pairing latent residuals with (i) block-declustered GPD fitting (replacing raw correlated exceedances N_u with independent cluster peaks scaled by the Ferro–Segers extremal index $\hat{\theta} \in (0, 1]$ [30], whose reciprocal $1/\hat{\theta}$ measures the average number of consecutive steps per tail burst), (ii) online conformal or rolling quantile updates gated by prediction-error variance under

TABLE III
OPERATIONAL LATENCY AND CALIBRATION-QUANTILE TRADE-OFFS: (A) TARGET-HORIZON SWEEP ($S \in \{1, 4, 16, 32, 64\}$, LAG $S-1$, n=5 STREAMS, 3 SEEDS) AND (B) DETECTION AT DEPLOYABLE NOMINAL CALIBRATION QUANTILES q^* (n=19 STREAMS; ER = EVENT RECALL, P = PRECISION, FA = FALSE ALARMS PER 1,000 STEPS).

(a) Target-Horizon and Score-Emission Sweep

		OpEntropy, causal			TS-JEPA, causal		TimesNet, causal		OpEntropy, buf	
S	Lag	FPR	ER	Δt	FPR	ER	FPR	ER	FPR	ER
1	0	0.005	0.729	76.3	0.004	0.524	0.025	0.895	0.005	0.729
4	3	0.007	0.624	179.3	0.006	0.395	0.031	0.910	0.020	0.481
16	15	0.003	0.257	200.3	0.003	0.257	0.061	0.700	0.043	0.391
32	31	0.011	0.219	397.9	0.003	0.276	0.098	0.657	0.057	0.376
64	63	0.020	0.410	209.8	0.004	0.200	0.213	0.614	0.099	0.367

(b) Detection at Matched Nominal Budgets q^*

		TS-JEPA			OpEntropy			TimesNet			TranAD		
Regime	q^*	ER	P	FA	ER	P	FA	ER	P	FA	ER	P	FA
Causal	0.001	0.493	0.285	88.1	0.556	0.228	93.7	0.590	0.219	222.4	0.533	0.156	209.6
Causal	0.050	0.730	0.251	177.3	0.752	0.259	202.8	0.618	0.221	262.3	0.602	0.159	269.5
Buffered	0.001	0.487	0.236	219.4	0.559	0.242	247.3	0.583	0.194	259.2	0.516	0.130	262.6
Buffered	0.050	0.582	0.233	283.7	0.654	0.268	317.6	0.612	0.196	287.3	0.596	0.145	295.3

TABLE IV
THRESHOLD CALIBRATOR COMPARISON AT DESIGN RISK q=10^{-3}: (A) STATIC RULES ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES), (B) CONTROLLED-DRIFT STRESS TEST (n=16), AND (C) THREE-STAGE REFITTED PERMUTATION DECOMPOSITION OF EXCESS CAUSAL FPR (n=18).

(a) Model	Calibrator Rule	Causal FPR	Overshoot $> q$	Buffered FPR	Point-F1 (buf)
TS-JEPA	Standard SPOT ($q = 10^{-3}$)	0.0171 (0.0151)	10%	0.0640 (0.0554)	0.1386
	SPOT + Kurtosis ($\beta = 0.167$)	0.0185 (0.0159)	10%	0.1630 (0.1397)	0.1498
	Empirical 99.5th Percentile	0.0254 (0.0229)	21%	0.0756 (0.0659)	0.1399
	Validation Maximum	0.0160 (0.0140)	7%	0.0658 (0.0568)	0.1385
OpEntropy	Standard SPOT ($q = 10^{-3}$)	0.0212 (0.0195)	14%	0.0725 (0.0638)	0.1573
	SPOT + Kurtosis ($\beta = 0.167$)	0.0209 (0.0188)	10%	0.1346 (0.1165)	0.1610
	Empirical 99.5th Percentile	0.0298 (0.0281)	27%	0.0882 (0.0783)	0.1633
	Validation Maximum	0.0187 (0.0170)	10%	0.0727 (0.0637)	0.1566
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.1795 (0.1611)	49%	0.2217 (0.1985)	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.1843 (0.1623)	46%	0.2281 (0.2011)	0.1352
	Empirical 99.5th Percentile	0.1914 (0.1752)	51%	0.2338 (0.2125)	0.1413
	Validation Maximum	0.1791 (0.1599)	48%	0.2263 (0.2032)	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	0.1601 (0.1423)	46%	0.2217 (0.1963)	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.1722 (0.1513)	43%	0.2453 (0.2151)	0.0940
	Empirical 99.5th Percentile	0.1712 (0.1544)	51%	0.2343 (0.2093)	0.0962
	Validation Maximum	0.1580 (0.1392)	44%	0.2270 (0.2009)	0.0952

(b) Injected condition	Static SPOT	Static empirical	Rolling empirical	Rolling SPOT
Stationary (no injection)	0.061	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.187	0.015	0.070
Variance scale 2.0	0.082	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.232	0.024	0.055
Continuous mean ramp	0.075	0.186	0.326	0.385

	Refitted Stage Causal FPR			Estimated Share of Excess ($\text{FPR}_{\text{chrono}} - q$)		
(c) Model	$\text{FPR}_{\text{chrono}}$	$\text{FPR}_{\text{block}}$	$\text{FPR}_{\text{point}}$	Inter-Block Shift (95% CI)	Serial Dep. (95% CI)	i.i.d. Tail
TS-JEPA	0.0121	0.0058	0.0019	56.6% (+.0063 [−.0054, +.0194])	35.3% (+.0039 [+.0001, +.0103])	8.1%
OpEntropy	0.0185	0.0073	0.0022	64.1% (+.0112 [−.0015, +.0336])	29.1% (+.0051 [−.0005, +.0146])	6.9%
TimesNet	0.0872	0.0204	0.0037	77.5% (+.0668 [+.0310, +.1082])	19.4% (+.0167 [+.0041, +.0338])	3.1%
TranAD	0.0741	0.0067	0.0032	92.3% (+.0674 [+.0354, +.1062])	4.7% (+.0035 [+.0002, +.0078])	3.0%

discrete regime shifts (Table IV(b)), or (iii) state-conditioned threshold tables indexed by telemetry operating mode. Three diagnostics in Table IV explain the static overshoot:

1) **GPD tail stability and quantile/length sensitivity:** On the 19 non-GHL streams at $u = p_{98}$, median calibration exceedance count is $N_u = 28$ causally (21 buffered) with 0% numerical solver fallback and low bootstrap threshold uncertainty ($\text{SE}(\tau)/\tau = 2.6\%$–$3.5\%$ causal). Causally, Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells (median $\xi = -0.27$) versus 34.4% for TimesNet and 54.4% for TranAD. Varying $u \in \{p_{95}, p_{98}, p_{99}\}$ leaves causal FPR unchanged (2.41%–2.42% for TS-JEPA), whereas halving calibration length (50% N_{cal}) raises causal FPR to 3.72% (TS-JEPA) and 4.13% (OpEntropy).
2) **Platykurtic collapse under total-threshold kurtosis scaling:** While total-threshold kurtosis scaling $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$ reduces buffered FPR on leptokurtic cells ($\gamma_2 \geq 0$, 4.49% → 1.11% for TS-JEPA), 64-step buffered smoothing renders 31.9% of TS-JEPA cells platykurtic ($\gamma_2 < 0$) with tight relative dispersion ($\sigma_{\text{cal}}/\mu_{\text{cal}} \approx 0.02$–$0.05$). Scaling total τ toward zero pulls τ_{eff} below the $u = p_{98}$ floor in 28.4% of cells, inflating overall buffered FPR from 6.40% to 16.30% (Table IV(a)). Scaling strictly the tail excess above u, $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$, prevents threshold collapse below u and holds 19-stream buffered FPR to 9.19% vs. 9.29% unadjusted (2.40% vs. 2.42% causal).
3) **Three-stage refitted permutation decomposition (Table IV(c)):** Simply permuting held-out evaluation scores against a fixed threshold cannot change the exceedance rate. Instead, our three-stage protocol pools the nominal calibration and evaluation vectors $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}] \in \mathbb{R}^{N_{\text{cal}}+N_{\text{hold}}}$ and *refits* the SPOT threshold ($u = p_{98}, q = 10^{-3}$) from scratch on each permuted calibration split: Stage 1 (chronological) fits τ_{chrono} on v_{cal} and evaluates $\text{FPR}_{\text{chrono}}$ on v_{hold}; Stage 2 ($L = 64$ block permutation) shuffles contiguous 64-step blocks across $\mathbf{v}_{\text{pool}}$, splits into permuted calibration (N_{cal}) and evaluation (N_{hold}) halves, *refits* τ_{block}, and evaluates $\text{FPR}_{\text{block}}$ (equalizing macro-block distributions while preserving intra-block autocorrelation); and Stage 3 (pointwise i.i.d. permutation) uniformly shuffles all points in $\mathbf{v}_{\text{pool}}$, splits into calibration and evaluation halves, *refits* τ_{point}, and evaluates $\text{FPR}_{\text{point}}$. Across the 18 streams with $N_{\text{cal}}, N_{\text{hold}} \geq 100$ (Table IV(c)), inter-block distribution shift ($\text{FPR}_{\text{chrono}} - \text{FPR}_{\text{block}}$) accounts for an estimated 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors), and serial dependence ($\text{FPR}_{\text{block}} - \text{FPR}_{\text{point}}$) for 29.1%–35.3%. Because latent predictors achieve $\text{FPR}_{\text{chrono}} = 0$ on 11 of 18 streams (concentrating stream-level variance onto the seven drifting streams so the TS-JEPA and OpEntropy inter-block 95% CIs in Table IV(c) span zero), these percentages represent estimated shares of aggregate excess risk rather than invariant per-stream causal proportions.

VIII. CONCLUSION AND FUTURE WORK

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) **Mechanism Isolation and Regularizer Equivalence:** Comparing 12 controlled backbones decouples score compression from representation learning: LayerNorm bottlenecks bound residual kurtosis and halve false alarms, whereas learned latent projections halve calibration-to-holdout distribution drift (D_{KS} : 0.376 → 0.209–0.214) and reduce causal false-positive rates to 0.9%–2.8%. On a fixed Koopman backbone, the spectral-entropy regularizer satisfies TOST equivalence down to $\Delta = \pm 0.010$ Point-F1, functioning as an operator conditioning constraint while the OpEntropy transition architecture shortens detection delay

relative to unconstrained TS-JEPA ($110.3 \rightarrow 79.5$ steps buffered; $127.0 \rightarrow 58.2$ causal).

2) **Relative False-Alarm Control vs. Detection Parity:** Latent predictive scoring lowers held-out false alarms three- to four-fold under buffered attribution and eight- to twenty-fold under strictly causal streaming, and nearly doubles causal event recall at stringent realized false-alarm budgets ($f^* = 0.01$), while overall Point-F1 and PR-AUC remain comparable to observation-space reconstructors.
3) **Calibration Drift and Operational Boundaries:** Static SPOT still overshoots its nominal 0.1% design risk by 15–$64\times$ out of sample, driven primarily by inter-block distribution shift (56.6%–64.1% of excess risk) and serial dependence (29.1%–35.3%). Future work should pair latent predictive residuals with declustered GPD fitting, online conformal quantile updates, and hybrid variance monitors to close the remaining calibration gap and cover quiet single-channel flatlines.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[8] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[11] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[12] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[13] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[14] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[15] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[16] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[17] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[18] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[19] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[20] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[21] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[22] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[23] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[24] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[25] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[26] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[27] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[28] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[29] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[30] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[31] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[32] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[33] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[34] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[35] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[36] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[37] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[38] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.